

Scalability & Future Plan

Date	15 July 2026
Team ID	XXXXXX
Project Name	AI Debt Settlement System
Maximum Marks	1 Mark

Scalability & Future Plan:

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses SQLite database, which is suitable only for small to medium-scale applications.	Limited scalability and concurrent user support.	High
2	AI recommendations depend on an external Google Gemini API service.	API rate limits and internet dependency may affect response time.	Medium
3	Supports only basic user authentication without advanced security features like Multi-Factor Authentication (MFA).	Increased security risks for production environments.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of concurrent users using a single FastAPI server.	Deploy using Docker and Kubernetes with Load Balancers (Nginx) to support thousands of concurrent users.
2	Data Storage	Uses SQLite for storing user and loan information.	Migrate to PostgreSQL or MySQL with cloud-managed databases (AWS RDS/Azure SQL).
3	Performance	Backend processes requests sequentially with limited caching.	Implement Redis caching, asynchronous processing, and API optimization for faster response times.
4	Security	JWT authentication with password hashing using bcrypt.	Introduce Multi-Factor Authentication (MFA), Role-Based Access Control (RBAC), HTTPS, and refresh tokens for enhanced security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on AWS/Azure with a production-ready PostgreSQL database.	1–2 Months	Improved reliability, scalability, and accessibility.
Phase 2	Introduce Role-Based Access Control (Admin, Financial Advisor, User).	2–3 Months	Better security and efficient management of multiple user roles.
Phase 3	Integrate Bank APIs and Credit Bureau APIs for automatic loan synchronization and credit score analysis.	3–5 Months	Eliminates manual data entry and provides real-time financial insights.
Phase 4	Develop Android and iOS mobile applications using React Native or Flutter.	5–6 Months	Increased accessibility and improved user experience across mobile devices.
Phase 5	Enhance AI capabilities using predictive analytics, repayment forecasting, and personalized financial planning.	6–9 Months	Smarter recommendations, improved financial decision-making, and better debt management outcomes.